Proposed Milestone-Based Stage Gates

Gate 1. MVP delivered and platform validated

Reporting period	November 2026 to September 2027
Verified at	Progress Report 1
Payment	Payment 2
Proposal basis	Phase 0 and Phase 1; Outputs 1, 2 and 3
Decision	Proceed to national and county deployment, or hold and remediate

Milestones

The MVP is a four-component deliverable due June 2027 and is an important milestone of First Gate.

1. RADAR MVP demonstrated and accepted by KNPHI by June 2027, comprising all four components: an agreed minimum event record for priority surveillance events; mapped and functioning interoperability and data flows for selected priority data sources; an integrated DMT-PHE decision-support workflow demonstrating signal ingestion, classification, escalation and human confirmation; and **initial technical specifications and interface requirements** for RADAR/DMT-PHE to GHEC-AI integration agreed jointly with the Sobek team.
2. Technical Product Profile and technical architecture signed off by WSU, University of Michigan/CGHE, KNPHI/DHA and the enterprise software development partner, covering deployment requirements, interoperability interfaces and security controls.
3. Dataset inventory and system mapping complete; priority system connectors and API/MCP interfaces built and functioning; outbreak user guides drafted across all four outbreak categories; national DMT-PHE deployment plan approved by KNPHI.
4. Eight-week parallel validation deployment of DMT-PHE completed at the national EOC, generating a concordance dataset against independent expert panel review.
5. Governance conditions met: KEMRI SERU amendment approved, WSU IRB approval extended, and data sharing agreements or MOUs executed for all priority data streams used in the MVP.
6. Baseline measurement completed and reported: detection-to-classification time, detection-to-escalation latency, and 7-1-7 compliance recorded for the national EOC before RADAR deployment.

Performance thresholds

- AI-expert concordance at or above 0.85 on the parallel validation dataset
- System Usability Scale score at or above 75 among national EOC users
- DMT-PHE documentation completeness at or above 80 percent across escalation events
- Zero adverse decision events attributable to the AI layer
- Minimum event record agreed with KNPHI and in operational use

- Baseline 7-1-7 and detection-to-classification measures documented and agreed with KNPHI

If the MVP is not accepted by 30 June 2027, the consortium should notify the foundation within 30 days with a remediation plan rather than waiting for the progress report. Gate 1 cannot be passed based on a partially delivered MVP given that Gate 2 depends on all four components.

Gate 2. Deployed, hardened and validated in operational use

Reporting period	October 2027 to September 2028
Verified at	Progress Report 2
Payment	Payment 3
Proposal basis	Phase 2; Outputs 4 and 5
Decision	Proceed to prospective evaluation and KNPHI transition, or hold scale-up

Milestones

1. RADAR and the AI-enabled DMT-PHE deployed and in routine use at the national EOC and five priority county EOCs, with site readiness assessments, ICT setup and staff training complete at every site.
2. Product hardening complete: scalability and load testing, cybersecurity and penetration testing, interface validation, and deployment within KNPHI/DHA-approved infrastructure, with test reports available to the foundation.
3. GHEC-AI escalation-to-response handoff validated in live workflow, confirming that only authorized, minimum-necessary operational information passes through the interface.
4. EOC staff cohort trained at national and county level, with training records and competency assessment results (Output 4).
5. Validated outbreak user guides, algorithms, workflow materials and training materials completed across all four outbreak categories (Output 5).
6. AI model retrained on national EOC operational data, with a documented benchmarking and recalibration procedure and a published error and override log.
7. Minimum event record published and adopted as a KNPHI standard.

Performance thresholds

- Dataset integration at or above 80 percent of the baseline inventory
- AI-expert concordance at or above 0.88 at every deployment site
- System Usability Scale score at or above 78 at every deployment site
- DMT-PHE documentation completeness at or above 85 percent across escalation events
- Zero adverse decision events attributable to the AI layer
- Median time from signal detection to structured DMT-PHE risk classification at or below 48 hours
- Measurable reduction against baseline in 7-1-7 detection and notification timelines at the national EOC and at a majority of county sites

Gate 3. Evaluated, institutionalized and transitioned

Reporting period	October 2028 to September 2029
Verified at	Final Report
Payment	None attached
Proposal basis	Phase 3; Output 6
Decision	Confirm grant closeout and inform any follow-on investment

Milestones

1. Prospective evaluation and endline assessment completed by WSU with KNPHI, using the baseline and endline comparison, 7-1-7 performance measures, system-use data, user surveys and key informant interviews described in the proposal.
2. KNPHI operational transition complete: the system runs on government infrastructure, administrative rights are transferred, and maintenance, support and security procedures are documented and handed over.
3. Recurrent budget pathway for RADAR and DMT-PHE documented and reflected in a KNPHI budget submission or departmental statement.
4. Scale-up evidence report and policy brief or departmental statement delivered (Output 6).
5. Two manuscripts documenting development and deployment submitted to peer-reviewed journals, with preprints posted.
6. Regional dissemination completed through Africa CDC epidemic intelligence networks and the WHO Preparedness Data Exchange.
7. Open access outputs released: de-identified or aggregated derivative datasets, metadata, analytic code, data dictionaries and model documentation, subject to Government of Kenya governance requirements.

Performance thresholds

- AI-expert concordance at or above 0.90
- System Usability Scale score at or above 80
- DMT-PHE documentation completeness at or above 90 percent
- False-negative rate below the pre-specified ceiling agreed at Gate 1
- Sustained reduction in detection-to-escalation latency against baseline
- Demonstrated improvement in 7-1-7 compliance against baseline at national and county sites

Notes on how these gates differ from the proposal

Three changes are worth flagging to the consortium before these gates are written into the grant agreement.

- The proposal's four phases are consolidated into three gates. Phase 0 (MVP, June 2027) and Phase 1 (platform preparation and parallel validation, September 2027) both fall inside the first reporting year, so they are combined into Gate 1 with the MVP retained as a hard internal deadline at month 8.
- Documentation completeness thresholds of 85 percent at Gate 2 and 90 percent at Gate 3, and the concordance threshold of 0.88 at Gate 2, are proposed here rather than taken from the proposal. They create a rising standard across the three gates and align the endpoint with the 90 percent and 0.90 targets already stated in the outcomes section. The remaining thresholds are drawn directly from the proposal's gated decisions.
- A 7-1-7 threshold has been added at Gate 2 and Gate 3. The proposal lists 7-1-7 compliance among its primary outcome measures but attaches it to only one gated decision. Without it, the gates test model accuracy and user acceptability and say nothing about whether outbreak detection and response actually improved.